Kyle Stanevich

kylestanevich@gmail.com kylestanevich.github.io (815) 895-1098

SUMMARY

Aerospace engineer with a background in satellites, high-energy projects, and systems engineering. Experienced in payload integration, hardware-in-the-loop testing, radiation analysis, and Python development. Eager to take on impactful programs from inception to completion alongside a driven team. Willing to travel and relocate.

EXPERIENCE

Rocket Lab (SolAero) – Albuquerque, NM **2024-present**

Senior R&D Engineer – Spacecraft Solar Cell Qualification and Reliability

- Led qualification efforts for emerging solar cell technologies, ensuring compliance with industry standards and customer requirements. Collaborated with development and production teams to resolve anomalies found while testing. Documented the results in qualification reports that were sent to customers.
- Planned and executed testing campaigns, including electrical characterization, irradiation, thermal shock, and humidity exposure. Consulted with subject matter experts and followed "test like you fly" principles.
- Developed schedules and coordinated resources to meet program milestones, collaborating with external labs to reduce costs and accelerate timelines.
- Established standardized testing guidelines for electrostatic discharge and atomic oxygen effects, enabling consistent evaluation processes.
- Extended Python data analysis and automation tools using AI agents (Claude via OpenCode).

Amazon (Project Kuiper) – Redmond, WA **2021-2023**

Systems and Test Engineer – Satellite Avionics

- Defined and managed requirements for unit acceptance testing, ensuring alignment with design specifications and flow down into test automation software.
- Led cost reduction initiatives for test equipment, optimizing across price, commonality, complexity, labor, and lead time to achieve significant savings while maintaining technical performance.
- Coordinated schedules and customer dependencies to track test system bring-up and early production testing.

Hardware Automation and Test Engineer – Propulsion System

- Led environmental and integration testing for the electric thruster unit. Designed, built, and operated test racks to validate flight hardware functionality during spacecraft environmental exposure tests.
- Designed hardware-in-the-loop (HIL) emulators to enable rapid development of the thruster system. Emulated endpoints included the anode, cathode, torque rods, valves, and heaters.
- Built a Python test framework that ran multi-day tests without human intervention, handling data recording, pass/fail analysis, and reporting. Developed custom python drivers to interface with the test rack instruments.

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Sandia National Laboratories – Albuquerque, NM **2020-2021**

Guidance, Navigation, and Control Engineer – Hypersonic Missile

- Implemented novel guidance laws in the CUDA C++ hypersonic missile simulator, utilizing Monte Carlo analysis to evaluate GNC performance.
- Developed a digital twin for the hypersonic missile avionics, modeling components and sensors in Simulink. Modularized components to enable rapid hardware-in-the-loop testing.
- Led fin and actuator sub-assembly testing, from test plan development through results analysis, and established the safety procedures governing the test setup.

General Atomics **2018-2020**

Railgun Development Engineer

- Analyzed in-bore projectile telemetry to refine railgun physics models, including a mechanism for contact-metal phase transition.
- Created a bore scanner using a laser sheet and camera to generate 3D cross-sections of the barrel, characterizing wear over time and cleaning effectiveness.
- Designed a projectile tracking system using stereo vision and high-speed IR cameras, achieving radar-like accuracy. Repurposed the system for shrapnel tracking during dispense testing, eliminating manual witness-card analysis.
- Automated the detection and tracking of imperfections during wound-barrel composite manufacturing, improving quality control and enabling continuous production.
- Calibrated a radar tracking system using RTK-GPS UAVs, enabling it to guide a simple, GPS-less projectile to its target in flight.

Aerospace Engineer – Proposal Development

- Formulated advanced control methods (PID, PTOC, and SMC) for the Next Generation Interceptor, evaluating performance using a MATLAB-based simulator.
- Developed a thermal fluid simulator for a high-power laser thermal management system, analyzing different layouts, fluids, and heat exchangers to maximize cooling runtime.
- Evaluated underwater buoyancy glider submarine designs using CFD and scale models testing. Results were used to define the set of possible mission profiles.

University of Illinois **2015-2018**

Researcher and Teaching Assistant

- Teaching assistant for the graduate spacecraft electric propulsion course, covering plasma physics and electric propulsion.
- Research assistant in the Helicon Injected Inertial Plasma Electrostatic Rocket (HIIPER) electric propulsion lab. Operated the thruster, rebuilt the helicon, and designed a micronewton thrust stand.
- Research assistant in the Center for Plasma-Material Interactions, working on tokamak and theta-pinch devices and a tin-lithium material study.

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EDUCATION

University of Illinois at Urbana-Champaign **GPA: 4.00** **2018**
Master of Science, Aerospace Engineering
electric propulsion, combustion, distributed and satellite control systems

University of Illinois at Urbana-Champaign **GPA: 3.97** **2017**
Bachelor of Science, Aerospace Engineering
control systems, CFD, systems engineering, UAVs, thermodynamics

KEY SKILLS

- Systems Engineering, Program Management, Payload Integration
- Spacecraft Systems Qualification and Testing
- Hardware-in-the-Loop (HIL) Testing
- Space Environment Effects (ESD, Atomic Oxygen, UV)
- Radiation Modeling and Semiconductor Degradation
- Test Automation and Data Analysis
- Python, C++, MATLAB, Simulink, SolidWorks, AI Agents
- Linux, Windows, Git, Jira, Confluence